Program 1: Introduction to Maven and Gradle

Installation :

Java (JDK) Installation :

Step 1: Type sudo apt install openjdk-11-jdk in the terminal

It will install the OpenJDK 17 (Java Development Kit) and accepts installation automatically.

```
shreya@shreyam:~$ sudo apt install openjdk-11-jdk
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openjdk-11-jdk is already the newest version (11.0.30+7-1ubuntu1~22.04).
0 upgraded, 0 newly installed, 0 to remove and 13 not upgraded.
```

Step 2: Installation can be confirmed by running the command java --version

```
shreya@ShreyaM:~$ java --version
openjdk 11.0.30 2026-01-20
OpenJDK Runtime Environment (build 11.0.30+7-post-Ubuntu-1ubuntu122.04)
OpenJDK 64-Bit Server VM (build 11.0.30+7-post-Ubuntu-1ubuntu122.04, mixed mode,
 sharing)
```

Maven Installation:

Step 1: Type the command sudo apt install maven in the terminal.

It will install Apache Maven from Ubuntu's package repository.

```
shreya@shreyam:~$ sudo apt install maven
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
maven is already the newest version (3.6.3-5).
0 upgraded, 0 newly installed, 0 to remove and 13 not upgraded.
shreya@shreyam:~$
```

Step 2: Installation can be confirmed by running the command mvn --version

```
shreya@ShreyaM:~$ mvn --version
Apache Maven 3.6.3
Maven home: /usr/share/maven
Java version: 11.0.30, vendor: Ubuntu, runtime: /usr/lib/jvm/java-11-openjdk-amd
64
Default locale: en_IN, platform encoding: UTF-8
OS name: "linux", version: "6.8.0-106-generic", arch: "amd64", family: "unix"
```

Gradle Installation:

Step 1: Type the command sudo apt install gradle in the terminal

It installs Gradle build automation tool from Ubuntu's package repository.

```
shreya@shreyam:~$ sudo apt install gradle
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
gradle is already the newest version (4.4.1-13).
0 upgraded, 0 newly installed, 0 to remove and 13 not upgraded.
```

Step 2: To check the proper installation of Gradle run the command gradle –version

```
shreya@ShreyaM:~$ gradle -v

------------------------------------------------------------
Gradle 4.4.1
------------------------------------------------------------
```